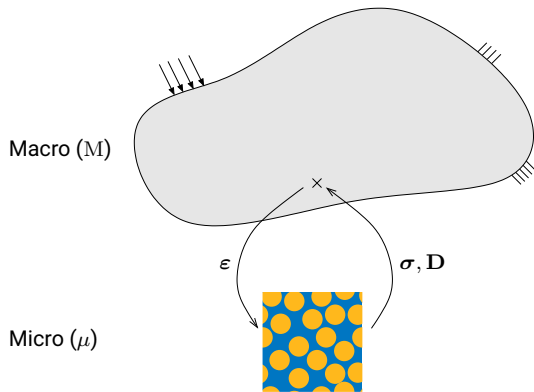


In fully coupled multiscale modeling (FE²) an RVE replaces the material model

Compute stress σ as the volume average in a deformed Representative Volume Element (RVE)



- Apply strain with periodic BCs

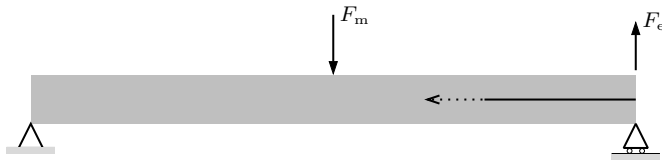
$$\mathbf{u}_{\mu}^{+} - \mathbf{u}_{\mu}^{-} = \boldsymbol{\epsilon}_{\text{M}} (\mathbf{x}^{+} - \mathbf{x}^{-})$$

- Compute macro-stress through averaging

$$\boldsymbol{\sigma}_{\text{M}} = \frac{1}{|\Omega_{\mu}|} \int_{\Omega_{\mu}} \boldsymbol{\sigma}_{\mu} \, d\Omega$$

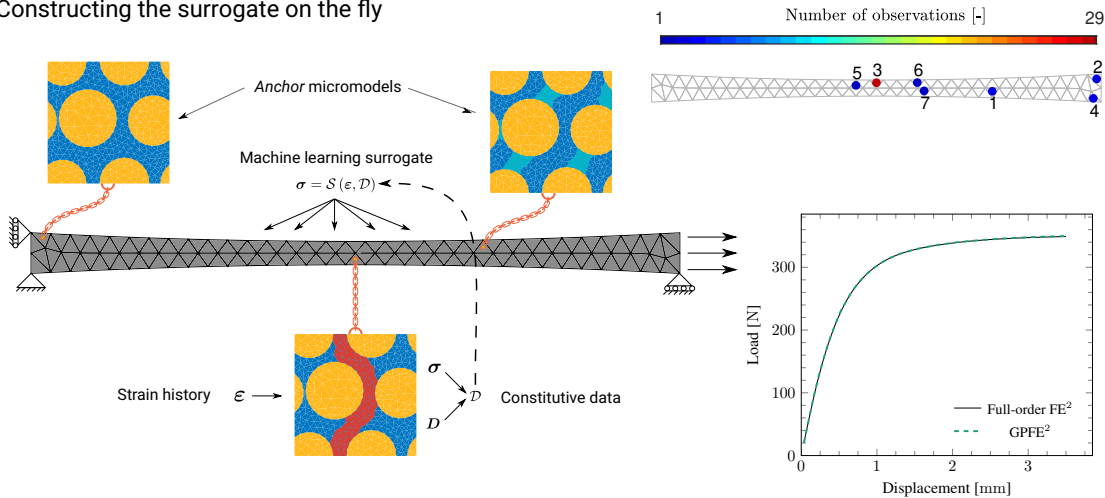
How far can we get on a real life example?

Multiscale modeling a mixed mode bending test

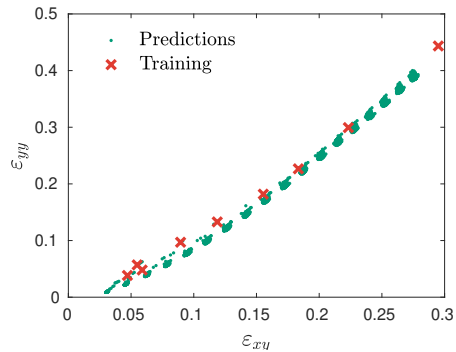
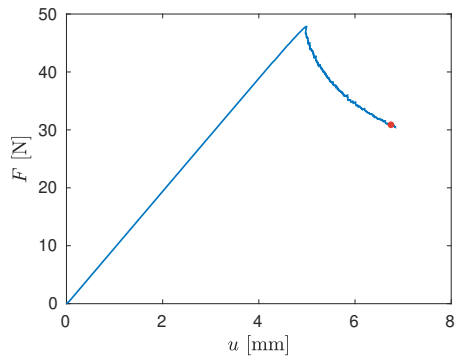


The Bayesian regression model allows for active learning

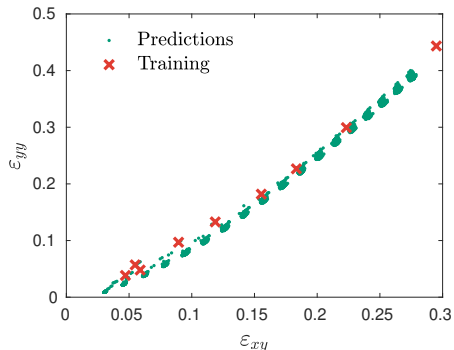
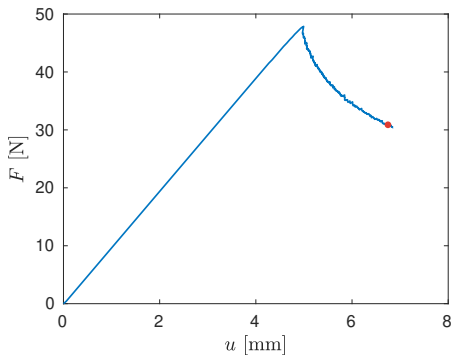
Constructing the surrogate on the fly



Many GP predictions are made with few micromodel runs



Many GP predictions are made with few micromodel runs



Cohesive GP reduction ratio: 1108 (80880 predictions / 73 full updates)

Bulk GP reduction ratio: 10676 (6448104 predictions / 604 full updates)